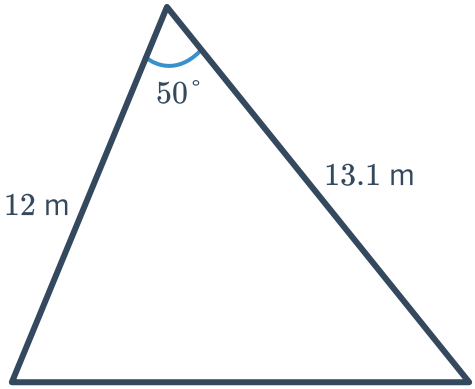
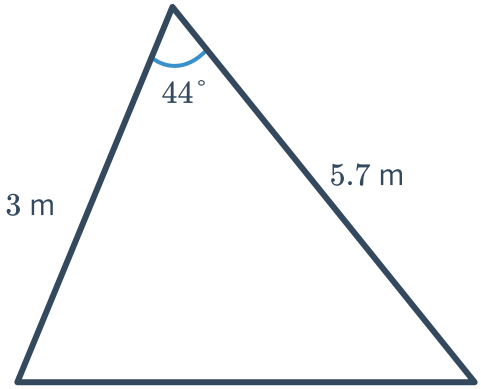
Area formula

1 Calculate the area of the following triangles to two decimal places:

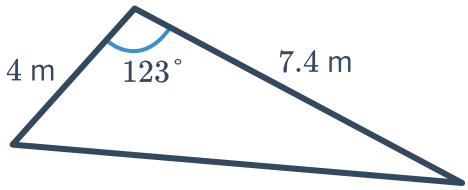
a



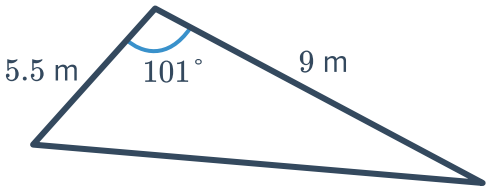
b



c



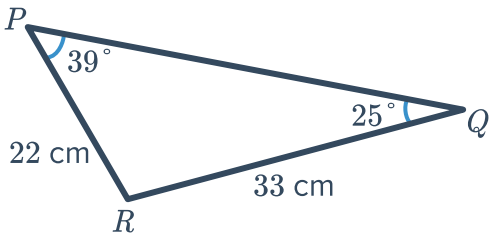
d



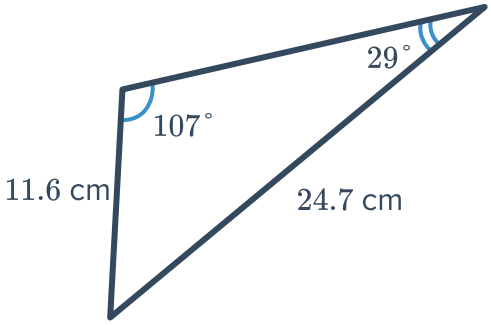
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2 Calculate the area of the following triangles, to the nearest square centimetre:

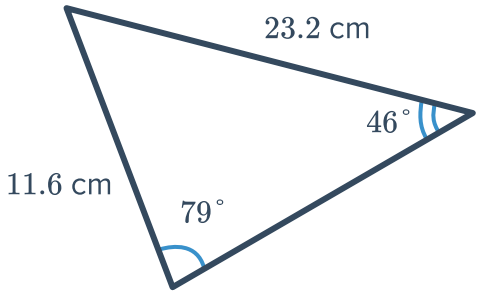
a



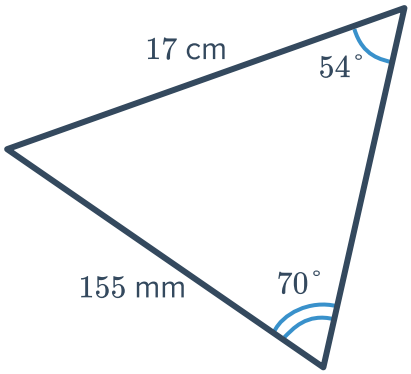
b



c



d



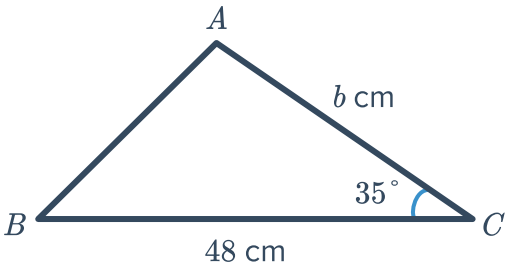
3 Find the area of the following triangles, to one decimal place:

a Sides lengths of $9.5\ \text{cm}$, $10\ \text{cm}$ with an included angle of 47°.

b Sides lengths of $18.4\ \text{cm}$, $20.5\ \text{cm}$ with an included angle of 99°.

Unknown sides and angles

4 The following triangle has an area of $520\ \text{cm}^2$. Find the length of side b. Round your answer to the nearest centimetre.



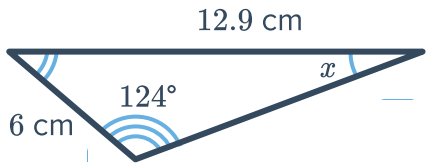
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5 For the following triangles, calculate the following:

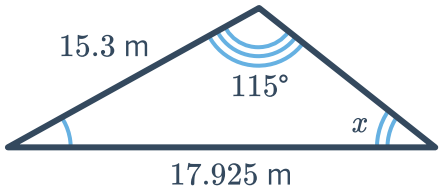
i The size of angle x, to the nearest degree.

ii The area of the triangle, to one decimal place.

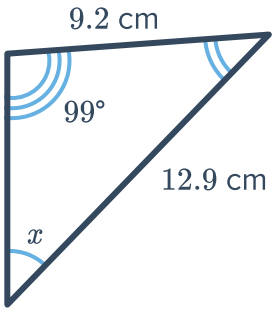
a



b



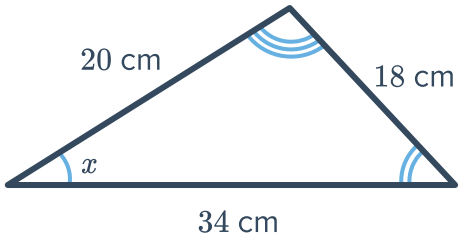
c



6 Consider the given triangle:

a Find the value of the angle x, to the nearest second.

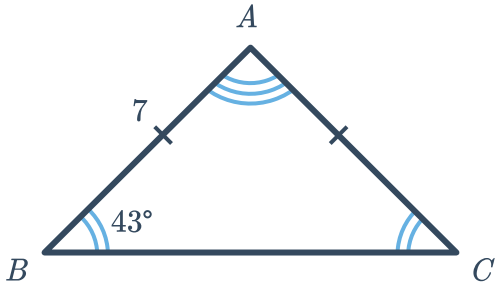
b Hence or otherwise, find the area of the triangle, to one decimal place.



7 Consider the triangle shown:

a Find the length of BC, to two decimal places.

b Hence or otherwise, find the area of the triangle, to two decimal places.

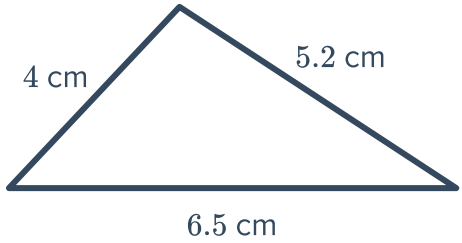


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8 Consider the given triangle:

a Calculate the size of the largest angle of the triangle. Round your answer to the nearest minute.

b Calculate the area of the triangle, to two decimal places.



9 Consider a triangle with side lengths of 8 cm , 4 cm and 6 cm.

a Find the value of the angle x opposite the side that is 8 cm long. Round your answer to two decimal places.

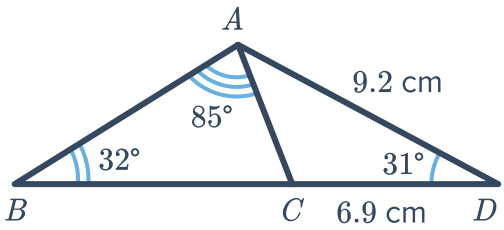
b Hence or otherwise, find the area of the triangle to two decimal places.

10 For the triangle shown, find the following, rounded to one decimal place:

a The length of AC.

b The area of $\triangle ACD$.

c The area of $\triangle ABC$.

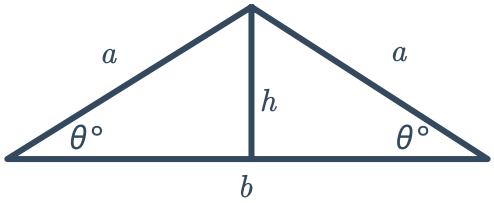


11 Consider the diagram of an isosceles triangle where h is the height perpendicular to base, b:

a Form an expression for h in terms of θ and a.

b Find b in terms of θ and a.

c Form an expression for the area A of the larger triangle, in terms of θ and a.



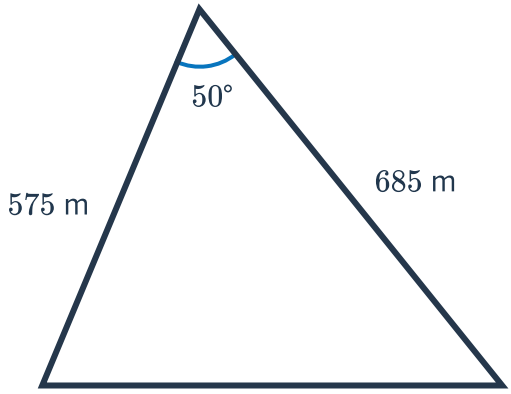
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Applications

12 A triangular paddock has measurements as shown in the diagram:

a Find the area of the paddock. Round your answer to the nearest square metre.

b State the area in hectares. Round your answer to two decimal places.

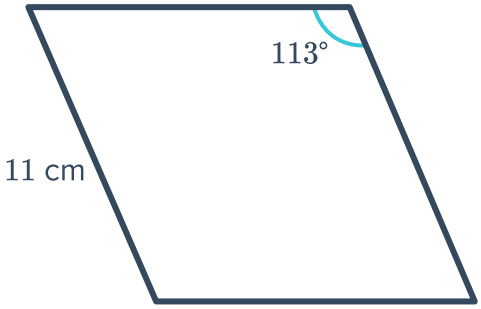


13 Find the area of each rhombus below. Round your answers to two decimal places.

a



b



14 A triangular-shaped field has sides of length $25\ \text{m}$, $29\ \text{m}$ and $36\ \text{m}$.

a Find the value of the angle x opposite the side that is $25\ \text{m}$ long. Round your answer to two decimal places.

b Hence or otherwise, find the area of the field to two decimal places.

c Bob has been hired to plough the field and to erect fencing around its perimeter. If he charges $\$3$ per square metre for ploughing and $\$7$ per metre for fencing, how much does he charge in total? Round your answer to two decimal places.

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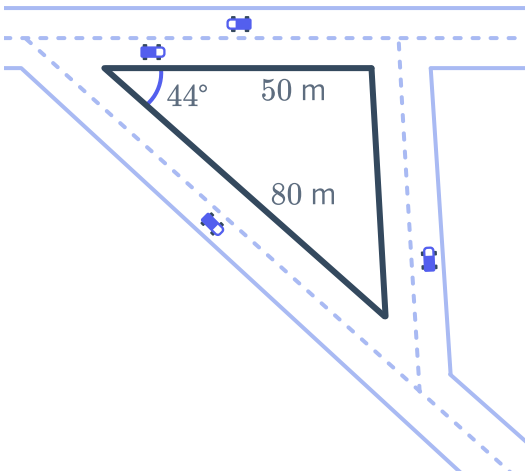
15 A triangle has sides in the ratio $13 : 14 : 15$ and a perimeter of 84 cm.

a Let the sides, in centimetres, be $13x$, $14x$ and $15x$. Solve for x.

b Find the value of the angle θ that is opposite the shortest side. Round your answer to two decimal places.

c Hence or otherwise, find the area of the triangle to two decimal places.

16 An industrial site in the shape of a triangle is to take up the space between where three roads intersect.
Calculate the area of the site, correct to two decimal places.

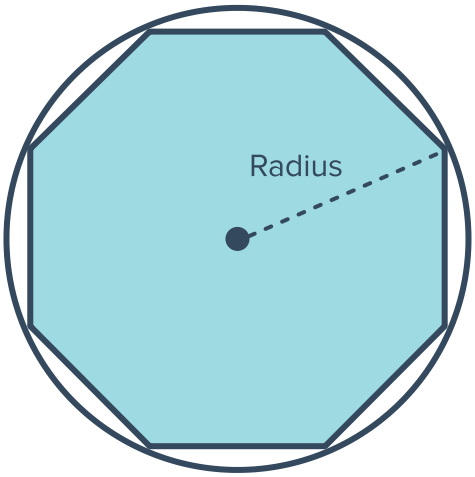


17 The Bermuda triangle is an area in the Atlantic Ocean where many planes and ships have mysteriously disappeared. Its vertices are at Bermuda, Miami and Puerto Rico.
Find the area taken up by the Bermuda Triangle. Round your answer to the nearest square kilometre.

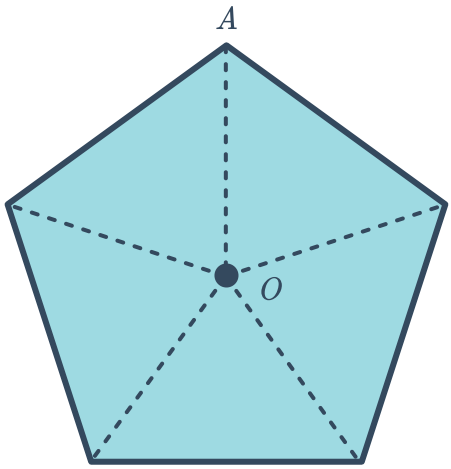


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18 An octagon is inscribed in a circle of radius $8\ \text{cm}$.
Find the area of the octagon leaving your answer in exact form.



19 A regular pentagonal garden plot has centre of symmetry O and an area of $86\ \text{m}^2$.
Find the distance OA.



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